

ISyE 6669 FALL 2011 – EXAM II

This exam is composed of three questions and totals 100 points. The point value for each question is listed next to the question. Budget your time accordingly. Read the entire exam before you begin. Start with the questions that you can solve quickly and correctly. You have ninety (90) minutes to complete the exam.

Partial credit will be liberally applied to incorrect answers that are well explained and well thought out. Be sure to precisely define all decision variables and explain all constraints if you formulate an LP model. In all questions, a little thought will be much more effective than a brute force attack. GOOD LUCK!

Question 1 (40 points) Consider the following LP:

$$\begin{aligned}
 \text{maximize } z = & \quad 3x_1 \quad + c_2x_2 \quad + c_3x_3 \\
 \text{subject to } & \quad A_{11}x_1 \quad + A_{12}x_2 \quad + A_{13}x_3 \leq 50 \\
 & \quad A_{21}x_1 \quad + A_{22}x_2 \quad + A_{23}x_3 \geq b_2 \\
 & \quad A_{31}x_1 \quad + A_{32}x_2 \quad + A_{33}x_3 = b_3 \\
 & \quad x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.
 \end{aligned} \tag{1}$$

and its associated optimal tableau

BV	z	x_1	x_2	x_3	s_1	e_2	a_2	a_3	rhs
z	1	1	0	0	1	0	M	$M+3$	$\bar{d}$
e_2	0	-3	0	0	1	1	-1	-2	15
x_3	0	0	0	1	1	0	0	-1	40
x_2	0	1	1	0	0	0	0	1	10

(2)

The variables s_1 , e_2 , a_2 and a_3 in the above tableau are the slack variable for the first constraint, the excess variable for the second constraint and the artificial variables for the second and third constraints of problem (1), respectively. Answer the following questions:

(a) **(8 points)** Show that the optimal basis A_B and its inverse are given by

$$A_B = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_B^{-1} = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) **(13 points)** Compute the coefficients c_2 , c_3 , A_{11} , A_{12} , A_{13} , A_{21} , A_{22} , A_{23} , A_{31} , A_{32} , A_{33} , b_2 , b_3 and $\bar{d}$.

- (c) **(13 points)** What is the range of values of b_1 for which the current set of basic variables remain optimal? Suppose b_1 is changed from its current value of 50 to 20. Using the dual simplex method, find the corresponding optimal BFS for the new problem.
- (d) **(6 points)** What is the range of values of c_1 for which the current set of basic variables remain optimal?

Solution. Let $A_{.4}$, $A_{.5}$, $A_{.6}$ and $A_{.7}$ (resp., $\bar{A}_{.4}$, $\bar{A}_{.5}$, $\bar{A}_{.6}$ and $\bar{A}_{.7}$) denote the columns of s_1 , e_2 , a_2 and a_3 in the original formulation (resp., optimal tableau).

a) Using the fact that

$$A_B^{-1}[A_{.4}, A_{.5}, A_{.7}] = [\bar{A}_{.4}, \bar{A}_{.5}, \bar{A}_{.7}]$$

or equivalently,

$$A_B^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix},$$

we conclude that

$$A_B^{-1} = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix},$$

and hence that

$$A_B = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix},$$

b) Using the fact that $\bar{b} = A_B^{-1}b$, or $b = A_B\bar{b}$, we conclude that

$$\begin{pmatrix} 50 \\ b_2 \\ b_3 \end{pmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 15 \\ 40 \\ 10 \end{pmatrix},$$

and hence that $b_2 = 15$ and $b_3 = 10$ **[3 pt]**.

Also, using the fact that $\bar{A}_{.j} = A_B^{-1}A_{.j}$ for all j , we conclude that

$$[A_{.1}, A_{.2}, A_{.3}] = A_B[\bar{A}_{.1}, \bar{A}_{.2}, \bar{A}_{.3}],$$

and hence that

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -3 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \quad \text{[4pt]}$$

Now, let $\pi^T = c_B^T A_B^{-1}$. Using the fact that $\bar{c}_j = \pi^T A_{.j} - c_j$ for every j , we conclude with $j = 4, 5, 7$ that

$$(1, 0, M+3) = (\pi_1, \pi_2, \pi_3) \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - (0, 0, -M),$$

and hence that

$$(\pi_1, \pi_2, \pi_3) = (1, 0, 3).$$

Hence, by the definition of π , we have

$$c_B^T = \pi^T A_B = (1, 0, 3) \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = (0, 1, 4)$$

Since $c_B^T = (0, c_3, c_2)$, it follows that $c_2 = 4$ and $c_3 = 1$. **[4 pt]**

Finally,

$$\bar{d} = c_B^T A_B^{-1} b = \pi^T b = (1, 0, 3) \begin{pmatrix} 50 \\ 15 \\ 10 \end{pmatrix} = 80. \quad \textbf{[2pt]}$$

c) The current set of basic variables remain optimal as long as

$$\bar{b} = A_B^{-1} b = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} b_1 \\ 15 \\ 10 \end{pmatrix} \geq 0,$$

or equivalently,

$$\begin{pmatrix} b_1 - 35 \\ b_1 - 10 \\ 10 \end{pmatrix} \geq 0.$$

Hence, the range on b_1 which guarantees optimality of the current set of basic variables is $b_1 \in [35, \infty)$. **[6 pt]**

If $b_1 = 20$, then the new $\bar{b}$ and $\bar{d}$ are

$$(\bar{b})_{\text{new}} = (-15, 10, 10)^T, \quad (\bar{d})_{\text{new}} = c_B^T (\bar{b})_{\text{new}} = (0, 1, 4)(-15, 10, 10)^T = 50,$$

and the associated canonical tableau is

BV	z	x_1	x_2	x_3	s_1	e_2	a_2	a_3	rhs
z	1	1	0	0	1	0	M	$M+3$	50
e_2	0	-3	0	0	1	1	-1	-2	-15
x_3	0	0	0	1	1	0	0	-1	10
x_2	0	1	1	0	0	0	0	1	10
ratio	X	1/3	X	X	X	X	M	$(M+3)/2$	
		$\uparrow$							

Performing an iteration of the dual simplex method, we obtain the following optimal tableau

<i>BV</i>	<i>z</i>	<i>x</i> ₁	<i>x</i> ₂	<i>x</i> ₃	<i>s</i> ₁	<i>e</i> ₂	<i>a</i> ₂	<i>a</i> ₃	<i>rhs</i>
<i>z</i>	1	0	0	0	4/3	1/3	<i>M</i> − 1/3	<i>M</i> + 7/3	45
<i>x</i> ₁	0	1	0	0	−1/3	−1/3	1/3	2/3	5
<i>x</i> ₃	0	0	0	1	1	0	0	−1	10
<i>x</i> ₂	0	0	1	0	1/3	1/3	−1/3	1/3	5

The new BFS is $(x_1, x_2, x_3, s_1, e_2) = (5, 5, 10, 0, 0)$. [7 pt]

d) The current set of basic variables remain optimal as long as

$$\bar{c}_1 = c_B^T A_B^{-1} A_{.1} - c_1 = \pi^T A_{.1} - c_1 = (1, 0, 3)(1, 2, 1)^T - c_1 = 4 - c_1 \geq 0,$$

or equivalently, $c_1 \in (-\infty, 4]$.

■

Question 2 (30 points) State University has three professors who each teach four courses per year. Each year, four sections of Marketing, Finance, and Production must be offered. At least one section of each class must be offered during each semester (Fall and Spring). Each professor's time preference and preference for teaching various courses are given in the table below.

	<i>Professor 1</i>	<i>Professor 2</i>	<i>Professor 3</i>
<i>Fall Preference</i>	3	5	4
<i>Spring Preference</i>	4	3	4
<i>Marketing</i>	6	4	5
<i>Finance</i>	5	6	4
<i>Production</i>	4	5	6

The total satisfaction a professor earns teaching a class is the sum of the semester satisfaction and the course satisfaction. Thus, professor 1 derives a satisfaction of $3 + 6 = 9$ from teaching marketing during the fall semester. Formulate a minimum cost network flow problem that can be used to assign professors to courses so as to maximize the total satisfaction of the three professors. (**Hint:** Consider a network having 3 nodes associated with each professor, 3 nodes associated with each course and 6 nodes associated with each pair (semester, course).)

Solution. See the following figure. Each "Professor" node has a net outflow of 4. Each "Course" node has a net outflow of -4. All other nodes have a net putflow of 0. Arcs from,

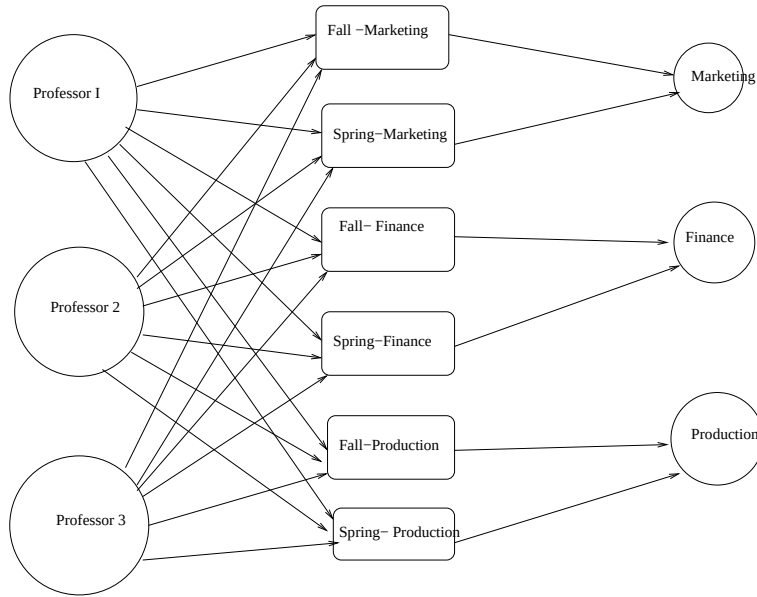


Figure 1: MCNFP formulation

say, Fall-Finance to Finance have a lower bound of 1 to ensure that the minimum number of courses are taught each semester. The cost for each arc from a "Professor" to a "season-course" node is $-(\text{reward associated with that teaching assignment})$. Thus Professor 1 to FallMarketing has a cost of $-(3+6)=-9$. ■

Question 3 (30 points) The following network depicts a partially solved minimum cost network flow problem (MCNFP) with labels on nodes indicating net supply and those on arcs showing unit cost, upper bound, and current flow. (The lower bound on the flow through every arc is assumed to be zero.)

- (a) **(7 points)** Is the given solution an extreme point (or BFS) for the MCNFP? [**YES:1pt**] If so, what are its basic arcs? [**Thick (1,2),(2,3),(3,5),(4,3),(5,6):2pts**] Specify the nonbasic arcs at lower bound [**Dotted (5,2):2pt**] and the ones at upper bound. [**Solid (1,3)(1,4),(4,6):2pt**]
- (b) **(7 points)** Consider now performing one iteration of the minimum cost network simplex method. Find the values of the multipliers, that is, the u_i 's associated with each node. [**$u_1 = 0, u_2 = -5, u_3 = -8, u_4 = -2, u_5 = -10, u_6 = -11$**]
- (c) **(8 points)** Determine the reduced costs for each of the nonbasic arcs

$$\bar{c}_{14} = -2, \bar{c}_{13} = 7, \bar{c}_{52} = -10, \bar{c}_{4,6} = 7 \text{ (4pts)}$$

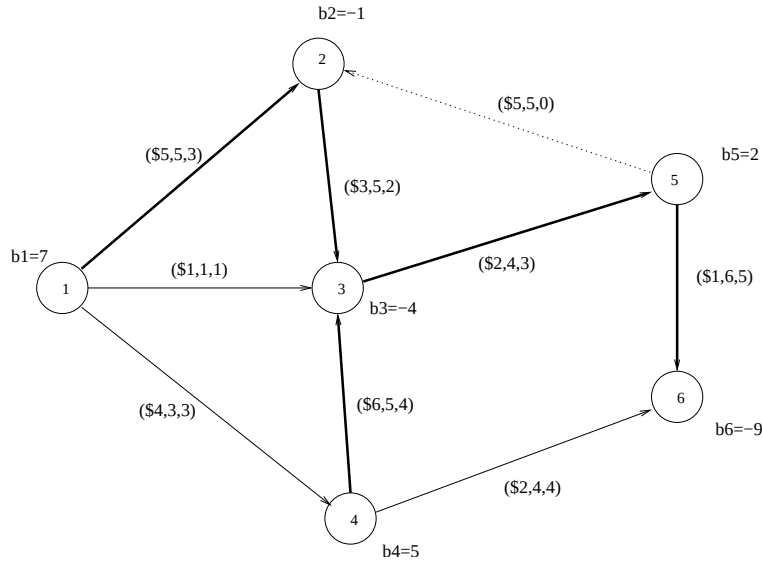


Figure 2: MCNFP Picture

and verify that the current solution is not optimal (3pts):

$\bar{c}_{14} = -2^*$ negative@upperbound nonbasic, violating optimality condition,
 $\bar{c}_{13} = 7$ positive@upperbound nonbasic, satisfying optimality condition,
 $\bar{c}_{52} = -10$ negative@lowerbound nonbasic, satisfying optimality condition,
 $\bar{c}_{4,6} = 7$ positive@upperbound nonbasic, satisfying optimality condition

Which nonbasic arc flow should be modified so that the current flow can be improved?
 [(1,4):1pt]

- (d) (8 points) Perform one pivot on the “best” available variable. Indicate the entering and leaving variables. [(1,4) entering, (1,2) leaving: 2pts]

Give the values of the new BFS obtained. The variables $x_{14}, x_{43}, x_{23}, x_{12}$ change to $x_{14} = 1, x_{43} = 2, x_{23} = 4, x_{12} = 5$ and the other variables remain unchanged. Variable x_{12} becomes nonbasic at upper bound and variable x_{14} becomes basic : 3pts]

By how much has the flow value improved? [\$4=reduced cost $\times\theta$: 3pts]